

Sri Sai Sumanth Duggina

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Summary

Software developer with over 2 years of experience building secure, scalable applications using Spring Boot, Java and AWS. Known for delivering innovative solutions through teamwork, adaptability, and continuous learning. Skilled in designing microservices architectures, developing RESTful APIs, and optimizing relational databases for performance and reliability.

Technical Skills

- **Languages & Frameworks:** Java 8/17, Spring Boot, Spring Data JPA, JavaScript, HTML5, CSS, Python
- **Databases:** PostgreSQL, MongoDB, Redis, MySQL
- **Cloud & Deployment:** AWS, Docker, Kubernetes
- **Messaging & Event Streaming:** Kafka, Amazon SNS, Amazon SQS
- **DevOps & CI/CD:** Jenkins, GitLab CI, Maven
- **Security:** JWT, OAuth2.0, AES-256
- **API & Protocols:** RESTful API, OpenAPI, Swagger
- **Testing & Tools:** Junit, Mockito, Selenium, Git, A/B Testing, Prometheus, Grafana, Postman

Professional Experience

Software Engineer, HSBC

05/2024 – Present | USA

- Worked on the Payment and Trade Information Management System, collaborating with cross-functional teams to gather requirements and implement features in an Agile environment, delivering over 20 features within sprints, ensuring alignment with business objectives.
- Guided backend development using Spring Boot, integrating SWIFT messaging for international trade and payment processing. Developed a RESTful API layer using OpenAPI/Swagger for design and documentation, handling trade data from over 50 sources and ensuring seamless communication for over 200,000 transactions daily. Utilized Postman for comprehensive API testing and validation.
- Designed and optimized the database using PostgreSQL, applying indexing and query optimization techniques, improving data retrieval speed by 40%. Incorporated security measures like OAuth 2.0, JWT, and OWASP best practices to ensure safe transactions and data integrity.
- Developed the frontend using React.js, focusing on user experience with responsive design. Implemented A/B testing, leading to a 15% increase in user engagement by optimizing the UI based on real user feedback and performance metrics.
- Integrated Grafana dashboards for observability, improving monitoring coverage and shortening mean time to recovery by 35%.
- Established automated CI/CD pipelines using Jenkins, GitLab CI, and Maven, leveraging Docker containers for environment consistency. Adopted Test-Driven Development (TDD) practices and integrated JUnit and Mockito for unit and integration testing, achieving 98% test coverage, reducing manual intervention by 30%, and accelerating the release cycle by 20%.
- Deployed the application on AWS, utilizing Elastic Beanstalk, EC2, S3, Lambda, and AWS API Gateway for scalability and Aurora for database management, ensuring high availability, fault tolerance, and seamless performance, handling over 50,000 concurrent transactions per day with zero downtime. Leveraged IAM for secure identity and access management.
- Augmented the system's scalability by extending the existing architecture with event-driven communication using Apache Kafka, enabling real-time data streaming across microservices and supporting over 10k active customers.

Associate Java Developer, LTIMINDTREE

01/2023 - 12/2023 | India

- Developed the CRM system using a microservices architecture with Spring Boot and Docker, collaborating in an Agile environment and leveraging Java & features. Implemented A/B testing for feature optimization, resulting in a 25% increase in user engagement and ensuring seamless integration with over 10 third-party APIs.
- Led frontend development using React.js and Redux for state management, designing dynamic user interfaces with Material-UI components. Integrated WebSocket for real-time notifications, improving customer interaction response time by 30%, and implemented responsive design for 95% cross-browser compatibility.
- Engineered the database schema using PostgreSQL and MongoDB, optimizing complex queries and introducing database indexing, partitioning, and caching with Redis. Eased data retrieval time by 40% and supported a user base of over 10k active customers.
- Developed robust backend services with Spring Boot, using Spring Data JPA for ORM and Hibernate for efficient data transactions. Applied RESTful APIs and GraphQL for flexible querying and used RabbitMQ for asynchronous communication between microservices.
- Enhanced security by integrating OAuth 2.0 for authorization, JWT for secure token management, and AES-256 encryption for data at rest. Enforced Role-Based Access Control (RBAC) for secure user authentication, ensuring compliance with GDPR and PCI DSS.
- Implemented Kubernetes deployments for container orchestration, reducing operational overhead by 30% and improving environment parity across stages. Managed infrastructure deployments using Terraform.

Education

Central Michigan University, Master of Science, Computer Science

01/2024 – 05/2025 | MI, USA

R.V.R & J.C College of Engineering, Bachelor of Technology, Information Technology

08/2019 – 05/2023 | Guntur, India

Certifications

AWS Solutions Architect Associate, Amazon Web Services, Credential ID: [link](#)

03/2025 – 03/ 2028